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Islamic Development Bank

Improving Women Access to Finance in the Rice Postproduction Value Chain



2024



Acknowledgement

This report is a unique example of what we can achieve when we work together. It exemplifies how effective partnership can drive impact within the development arena. It is indeed pleasing that the Islamic Development Bank (IsDB) secured a grant of USD 11.25 Million from the Women Entrepreneurs Finance Initiative (We-Fi) to implement the Program on Empowering West African Women Small and Medium Enterprises in Rice Value Chains (EWASME) in four West African countries - Guinea, Niger, Senegal and Sierra Leone.

Within the framework of the EWASME Program, the Economic and Social Infrastructure Department and the Resilience and Climate Action Department of the Global Practices and Partnership Directorate of the Islamic Development Bank developed this report.

We are grateful to - Patrick M. Kormawa of Belvedere SL. Ltd., Sierra Leone for working diligently on preparing this report. The content draws from other reports on the study "Empowering West African Women Small and Medium Enterprises in Rice Value Chains", as well as the Guinea Report by Amadou Macky Dioum, the Niger Report by Hama Seyni, the Senegal Report by Saboury Ndiaye, and the Sierra Leone Report by Patrick M. Kormawa.

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The supervision, comments and guidance provided by Nizar Zaied - Manager of Agriculture, Water Resources, and Rural Development; Idrissa Dia - Director, Economic and Social Infrastructure Department; and Syed Hussein Quadri, Director - Resilience and Climate Action Department were critical in ensuring the quality of the output as you will notice while reading through the pages of this report.

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LIST OF ABBREVIATIONS

ECOWAS	Economic Commission of West African States
EWASME	Empowering West African Women Small and Medium Enterprises Program
FAO	Food and Agricultural Organization of the United Nations
FAOSTATS	Food and Agricultural Organization of the United Nations Statistics
IFC	International Finance Cooperation
IsDB	Islamic Development Bank
SMEs	Small and Medium Enterprises
VSEs	Very Small Enterprises
WDI	World Development Index
We-Fi	Women Entrepreneurs Finance Initiative
WSMEs	Women Small and Medium Enterprises
WVSEs	Women Very Small Enterprises



FOREWORD



Rice is one of the most stable foods for a large part of the global population and is extremely vital for food security for at least half of the World's population. It is expected that rice production will increase to 567 million tonnes (Mt) by 2030 according to projections by the Organization for Economic Cooperation and Development, and Food and Agriculture Organization.

The West African region is the largest producer of rice in the African continent producing a little over 10 million tons annually. Rice production provides livelihood to West African rice farmers, creating jobs and serving as a source of income for millions of smallholder farmers. Thus, the rice sector represents a pathway out of poverty and food insecurity, particularly for women, thus contributing significantly to sustainable development.



To bridge the existing gaps in food security, particularly in rice production, we must harness the untapped opportunities available, especially among entrepreneurs, with a focus on youth and women entrepreneurs who invest a lot of energy in food production in Africa. According to the African Development Bank, there is currently about USD42 billion funding gap for women entrepreneurs in Africa. However, it is projected that Africa could gain USD316 billion in Gross Domestic Product (GDP) by 2025 if the gender gap is bridged. Thus, supporting women entrepreneurs in any country in Africa is a strategic approach towards bolstering the economy.

I am delighted that the Islamic Development Bank is an implementing partner of the Women Entrepreneurs Finance Initiative (We-Fi) a Financial Intermediary Fund (FIF) established in 2017 to address financial and non-financial constraints of women entrepreneurs.

The Bank was granted support from We-Fi to implement the Empowering West African Women Small and Medium Enterprises in Rice Value Chains (EWASME) Program in Guinea, Niger, Senegal and Sierra Leone. The program aims to upgrade and advance women owned/led small and medium enterprises and women owned/led very small enterprises, particularly at the post-production stage, through capacity development, grant matching, and facilitating increased access to local and regional markets. It complements the existing Regional Rice Value Chain Program currently under implementation by the Islamic Development Bank in Guinea, Niger, Senegal and Sierra Leone.

The purpose of this report on Women Very Small Enterprises and Medium and Small Enterprises in the Rice Postproduction Value Chain, is to provide data and information. This data and information could be used to upgrade and advance women owned or women led very small enterprises, and women small and medium enterprises in rice post-production activities. The report focuses attention on the binding constraints that women entrepreneurs in rice post-production face and identified applicable solutions to the challenges women entrepreneurs encounter with respect to access to finance.

The report offers additional insights into the constraints and opportunities faced by women entrepreneurs, including very small, and small, and medium entrepreneurs in the rice post-production activities. It is hoped that it will advance knowledge and increase understanding about the rice postproduction activities especially with respect to finance.



It is further expected that this report will contribute to improving policies, programmes, and practices. It is intended to serve as a reference and information tool for policy makers, and the general public interested in rice production and food security.

The development of this report included the review and synthesizing of four country reports (Guinea, Niger, Senegal and Sierra Leone) based on a study commissioned by the Islamic Development Bank. Data and information from the country reports were complemented by desk research to enrich the content of this thematic report. The desk research related work on Women owned/led SMEs in general and in West Africa were reviewed to distil common threads that run through women owned/led VSE and SMEs.

I am confident that this report will add to the pool of information and knowledge resources that will contribute in enhancing the quality of intervention in our projects.

Dr. Issa Faye

Dr. Issa Faye,
Director General, Global Practice
and Partnerships

EXECUTIVE SUMMARY

Women entrepreneurs play important roles in the economy of countries in West Africa. However, despite their contribution and role in growing the economy, women entrepreneurs are confronted by distinct bottlenecks that inhibit the growth of their businesses. More women tend to operate in the informal sector and make a significant contribution to the agricultural sector. Based on data from the World Bank[1] (ILO estimates), the average percentage of the female population aged between 15 and 64 (labor participation rate) for the four study countries was 50% in 2022, ranging from 39% in Senegal, 44% in Guinea, 53% in Sierra Leone, and 64% in Niger.



Rice is a strategic crop and commodity: West Africa is the largest rice producing region in Africa, producing a little over 10 million tons annually. Women are involved as entrepreneurs at each segment of the rice value chain, especially in the postproduction processes. Rice production provides livelihoods to West African rice farmers providing jobs and as a source of income for millions of smallholder farmers. Thus, the rice sector represents a pathway out of poverty and food insecurity in the region as rice availability and prices have become major determinants of the welfare of the poorest sections of African consumers. The gross production value of rice produced in West Africa has been increasing consistently and hit almost USD 10 billion in 2022.

Women Very Small-Enterprises (WVSEs) and Women Small and Medium Enterprises (WSMEs) play important and distinct roles in the rice postproduction value chain in West Africa. Postproduction activities of WVSE and WSMEs are mostly carried out using basic technologies. This is further exacerbated by limited incentives for improving access to postproduction technologies, leading to poor quality milled rice and low prices and profitability of domestic rice. Along the rice postproduction value chain, the number of WVSEs and WSMEs diminishes as value and technology use increases.

Financial and non-financial constraints: WVSE and WSMEs in the rice postharvest value chain face both financial and non-financial constraints on their businesses. The most significant financial constraints are; (i) limited and poor access to finance; (ii) high cost of finance; (iii) cumbersome application procedures; and (iv) exclusion from financial and banking services due to their sizes, physical location, and distance from credit sources. Non-financial constraints are due to factors related to the



1. https://genderdata.worldbank.org/en/indicator/sl-tlf-acti-zs



47% of women owned firms have more than 50% female workers, while 74% of men owned firms have less than 50% female employees. With respect to climate change, women entrepreneurs also tend to support the transition to green growth.

direct operations of rice postproduction processes. These are as follows: (i) limited access to good quality milling machines, equipment, or services; (ii) poor access or absence of rice-drying floors or facilities; (iii) unavailability of labor-saving threshing equipment; (iv) poor storage facilities for paddy and milled rice; (v) poor knowledge and access to postproduction machines and equipment; (vi) shortage of labor; (vii) high economic cost on businesses; and (viii) limited capacity of financial institutions to provide agricultural lending and Islamic financial services.

Importance of financing women entrepreneurs: Financing the women entrepreneurs contributes directly to women's economic empowerment and significantly boosting the economy. Studies have shown that women-led firms are more likely to have sound management practices in place. Women-led enterprises bridge the existing gender gap, and thus, move towards the universal goal of achieving gender parity, enabling women to leverage their potential in the transformation of the economy and society at large. Women-led firms generate positive spillovers with their greater use of training courses (40% of women-led firms compared to 34% of men-led firms), contributing to the human capital of the company and empowering employees beyond the firm. Women also attract and retain more women, positively affecting female employment and reducing gender disparities, while promoting more inclusive growth. Notably, 47% of women-owned firms have more than 50% female workers, while 74% of men-owned firms have less than 50% female employees. With respect to climate change, women entrepreneurs also tend to support the transition to green growth.

Providing value chain finance to women enterprises: The financing needs of WVSEs and WSMEs differ across rice postproduction value chain. The financial needs vary depending on the level and sophistication of technology used. They range from USD 400 to USD 1,000 per hectare for harvesting and transfer. For threshing and winnowing, an investment outlay of USD 400 to USD 3,000 is needed, whereas for aggregation and storage, the estimated financial need ranges between USD 800 and USD 46,000. In the case of purchasing and installing a modern and energy-efficient facility and construction of a drying floor, an amount of USD 2,500 to USD 10,000 will be required. The purchase of a modern rice mill (2-5t/ha), installation, and training are estimated at USD 45,000,

while it will cost about USD 500 if a service provider mills the rice. Construction of a modern retail store and outlet is estimated to cost USD 45,000 to USD 50,000, including land cost.

Alleviating Financial Constraints: Alleviating financial constraints of WVSEs and WSMEs should take into consideration the taxonomy and context specific to each of the two categories. Thus, addressing the financial constraints should include: (i) provision of innovative financial products, such as blended finance; (ii) building the capacity of financial institutions to support WVSE and WSMEs; and (iii) improving women access to property and collaterals. In the case of addressing non-financial constraints, the key actions are: (i) develop country-specific diagnostics of WVSEs and WSMEs; (ii) support access of women to motorized machines and equipment; and (iii) build financial institutions specialized in Islamic banking services.

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01

CONTEXT AND **BACKGROUND**

Women-led businesses play an essential role in creating jobs, raising standards of living, and boosting inclusive growth. However, women entrepreneurs, continue to face constraints in accessing finance, capital, and technology. Also, women entrepreneurs experience legal, regulatory, and policy barriers, including difficulties in registering businesses, and changing social and cultural norms. The Women Entrepreneurs Finance Initiative (We-Fi) recognizes these problems, and therefore supports women entrepreneurs to overcome these constraints by scaling up access to financial products and services, building capacity, expanding networks, offering mentorship, and providing opportunities to link with domestic and global markets. In practice, We-Fi supports innovative programs in cooperation with major multilateral development banks to address the constraints binding women entrepreneurs. In particular, We-Fi provides support to expand solutions that enhance access to finance, markets, and technology, and strengthen skills, networks, and policies for women entrepreneurs.



The estimated population of West Africa as of 2024 is approximately 456.25 million people, out of which 226.20 million are female, representing 49.58% of the total population[2]. A breakdown of the population by country shows that about five countries in West Africa have more females than men (Nzeakor, 2023) and contribute significantly to employment and entrepreneurship. However, more women than men tend to operate in the informal sector and make a significant contribution to the economy and to the agricultural sector in particular. With respect to the agricultural sector in West Africa, women account for 80% of employment in processing, 70% of distribution, and almost 90% of sales of ready-to-consume agricultural products (Allen et al., 2018).

Women also play important roles as entrepreneurs and in cross-border trade in West Africa. However, despite the important roles they play and their contribution to the economy, women entrepreneurs are confronted with distinct bottlenecks that inhibit the growth of their businesses, including limited access to finance, infrastructure, and technology. Over the years, West African governments and development partners have recognized the importance of women entrepreneurs in socioeconomic development and are putting in place support mechanisms to alleviate the constraints they face. One such initiative is the "Empowering West African Women SMEs in the Rice Value Chains" (EWASME) program that is being implemented by IsDB with financial support from the We-Fi.

The EWASME program aims to support women entrepreneurs in four major rice-producing countries in West Africa (Guinea, Niger, Senegal and Sierra Leone) to better engage in the rice value chain, particularly in the postproduction stage. The program addresses three main areas: (i) financial constraints; (ii) non-financial constraints; and (iii) poor links between women entrepreneurs and existing private sector-led value chains that could address constraints in input and output markets. This report was prepared with assistance from IsDB to provide knowledge to help key stakeholders in supporting WVSE and WSME value-chain finance initiatives, in addition to the implementation of EWASME.



2 World Population Prospects (2024 Revision) - United Nations population estimates and projections. https://population.un.org/wpp/

1.1 Definitions of Key Concepts

Rice Value Chain: The term "rice value chain" describes the full range of value-adding activities that firms/enterprises, farmers and workers carry out to bring the product (rice) from its conception to its end use and beyond. The benefits of supplying specific value chains vary dramatically, depending on how these chains are governed; whether suppliers receive inputs, knowledge, and ancillary services as part of their engagement; and the targeted end market.

Rice Postproduction System: This consists of a set of operations that cover the phases from harvest to consumption. An efficient postproduction system aims to minimize losses and maintain the quality of the crop (or product) until it reaches the final consumer.

Rice Postproduction Process: This includes a series of processes and handling techniques that are part of rice cultivation. They are applied from the harvest from the field for the purposes of transformation into a form, condition, or composition that adds value, makes it storable or prolongs its shelf-life, and makes it consumable or usable in other value-added products (see Figure 1).

Transport: Following its harvest, the rice is transferred from the field to a location for further handling and processing.

Threshing: Is the process of removing the paddy (husk rice) from the panicle for further processing. There are three methods for threshing rice: **rubbing**, **impact**, and **stripping**. During threshing, the paddy is detached from the panicle. Women carry out this manual process by trampling the panicles. Where mechanical threshers are available, the impact method is used.

Drying: At the time of harvest, the moisture content of the grains ranges from 24% to 26%. Drying reduces the moisture content of the paddy to 12-14% for safe storage and milling, or at most 18% for temporary storage of two weeks when it is not possible to dry any faster.

Paddy Cleaning/Winnowing: This involves the separation of undesirable material, such as weed seeds, straw, chaff, panicle stems, empty grains, immature and damaged grains, sand, rocks, stone, dust, plastic, and even metal and glass particles.

Storage: This involves the safekeeping of paddy or milled rice for the seed and food, respectively. At the farm household level, storage is essential for food security or as a commodity bank for conversion into cash when required.

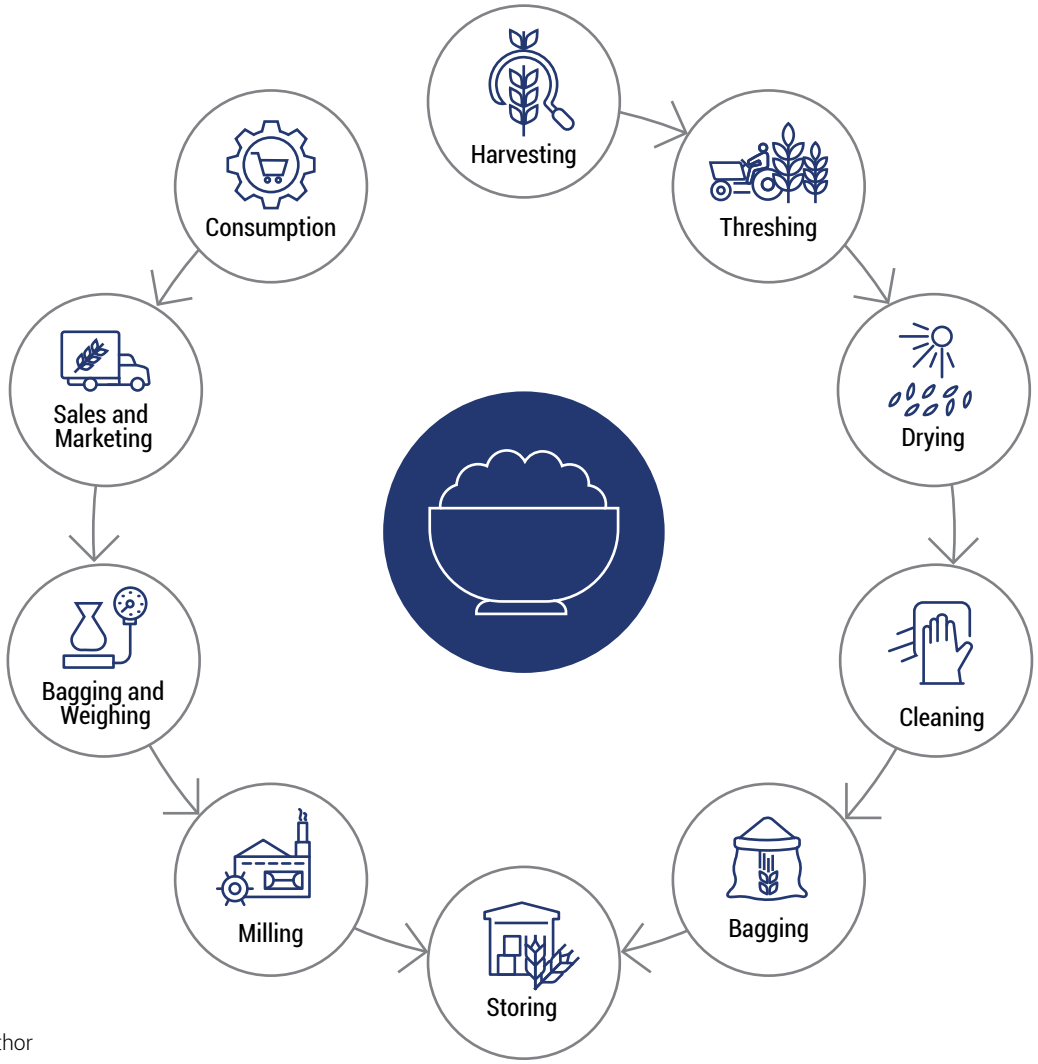
Parboiling: Paddy rice to be milled as parboiled rice is completely soaked in containers (metal drums, large pots, etc.) between 12 and 24 hours, usually by women. The water is drained to about 20%, the container is covered, and the paddy rice is steamed over fire.

Milling: This operation involves the separation of the husk and the bran (polishing) to produce the edible portion for consumption. Rice mills vary from a manually operated beam hammer pounder or mortar and pestle to the sophisticated machines used in big commercial or government installations. In rural areas without energy, the beam hammer pounder or the mortar and pestle are used by farmers. Mortar and pestle is the most common rice milling method by rural women. Women entrepreneurs who purchase paddy rice transport the paddy to commercial mills.

Women-owned Enterprises in Rice Postproduction: The definition of women-owned enterprise is provided in Box 1. The definition considers the ownership structure (owned by a majority women) and level of investment (51% or more). It also considers the proportion (20% or more) of women in senior-management positions.

Postproduction Value Chain Finance: This is funding provided to or by a value-chain actor to increase value-chain growth and competitiveness. In practice, value chain finance can be provided by a bank; a buyer; or an input, machine, or equipment supplier. Value chain financing allows enterprises or firms to operate and transact with others and to upgrade. It involves providing financial services and products to one or more points in a value chain to reduce the risk and cost of financing and to increase the efficiency and competitiveness of the value chain.

Figure 1: Rice Post-production processes



Source: Author

Box 1: Definition of Women-owned Enterprise.

Woman Owned Enterprise

An enterprise qualifies as a woman-owned enterprise if it meets the following criteria:

Criteria 1:

The enterprise is owned by 51% or more by a woman or women. In other words, the 51% of the investment or more is owned by a woman or women.

Or

Criteria 2:

The enterprise is owned by 20% or more by a woman or women and has one (1) woman or more serving as Chief Executive Officer (CEO), Chief Operating Officer (COO), President, or Vice President.

AND

Has 30% or more women serving as Board of Directors, where a board exists.

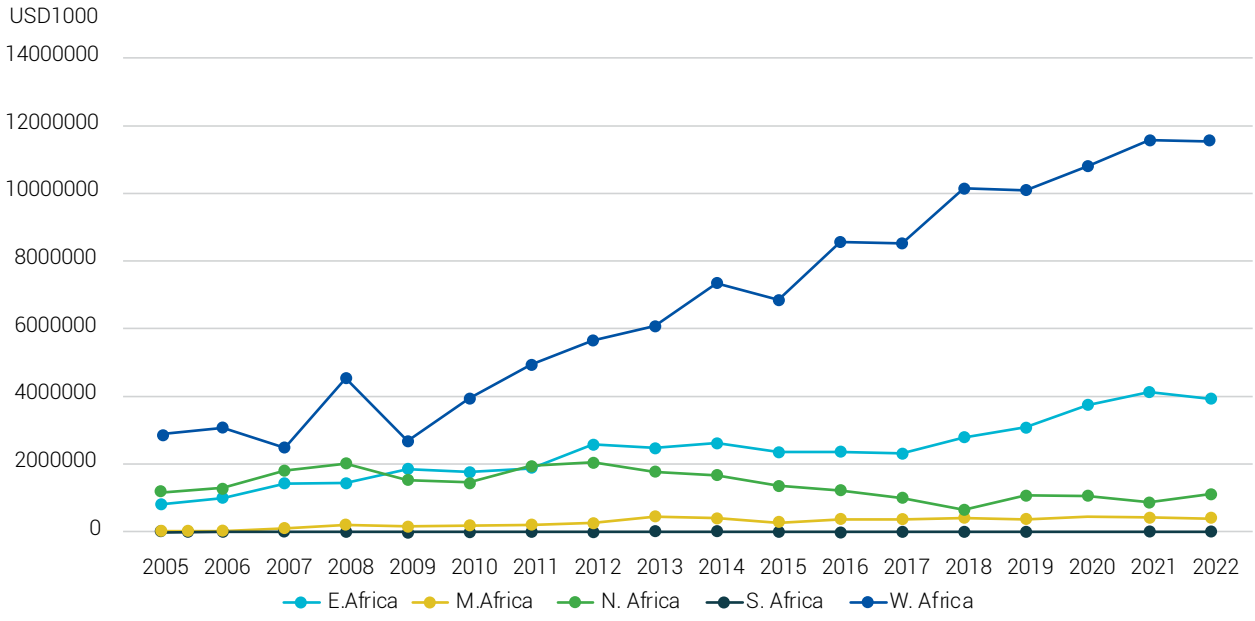
1.2 Rice in West Africa



West Africa is the largest rice-producing region in the continent, producing a little over 10 million tons annually. Rice production provides millions of smallholder farmers in the region with jobs, sources of income, and livelihoods. Thus, the rice sector provides an exit from poverty and food insecurity in the region, as rice availability and prices have become major determinants of the welfare of the poorest segments of African consumers. Figure 2 shows the gross production value of rice in each of the five regions of Africa. In West Africa, it has been increasing consistently, hitting almost USD 10 billion in 2022. East Africa is the next important region in terms of gross production value, with about USD 4 billion in the same year. While rice is produced in the other regions, their gross production value remains at USD 2 billion and tends to be declining.

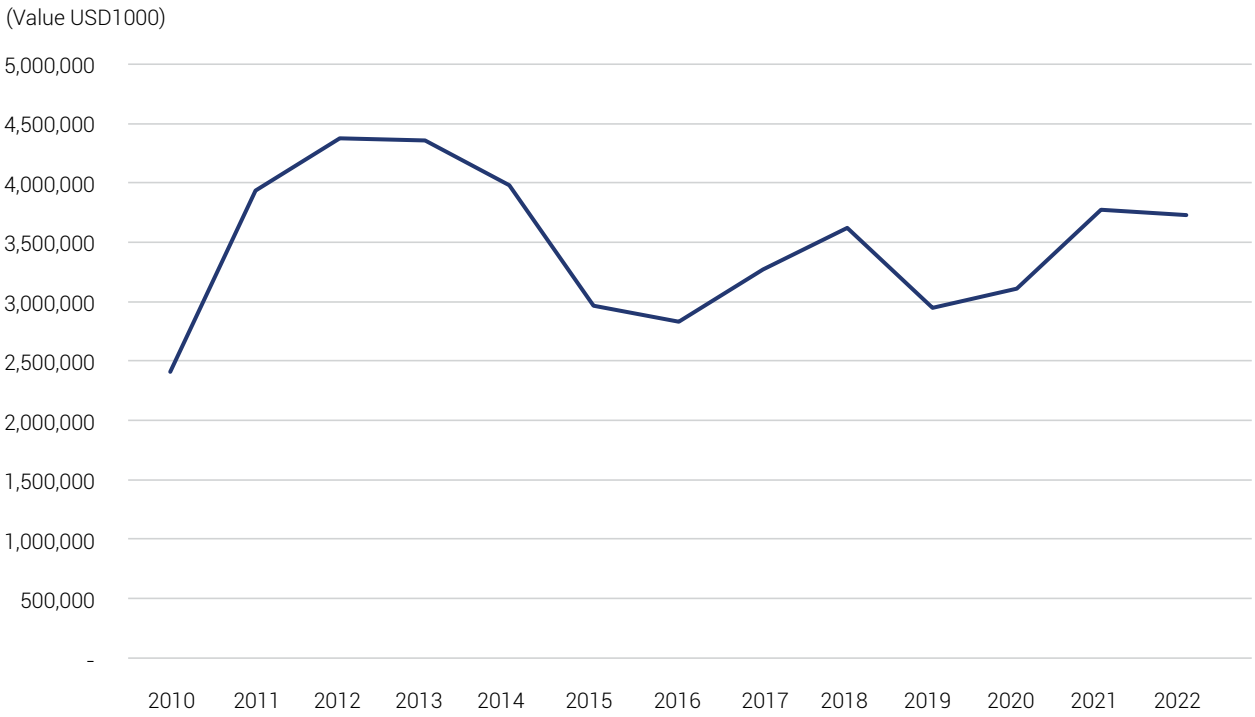
Although rice production and gross production value have been increasing in West Africa over the past two decades, consumption has been increasing faster than production. This increase has been driven mainly by the growing population, urbanization, and changes in consumer preferences. According to Africa Rice, the demand for rice is growing at more than 6% per year, which is faster than for any other food staple in Sub-Saharan Africa.

Figure 2: Gross Production Value 2005-2022 (current USD 1000)



Source: Author, based on FAOSTAT.

Figure 3: Rice Import Value in West Africa 2010-2022



Source: Author, based on FAOSTATS

Data from FAO shows that Guinea, Guinea-Bissau, Liberia and Sierra Leone have the highest per capita rice consumption (more than 90 kg per capita per year), followed by Senegal, Benin, Côte d'Ivoire, the Gambia and Mali (with more than 50 kg per capita per year) (FAOSTAT, 2024). According to FAO, rice consumption in West Africa is expected to grow by 70% to 24 million metric tons by 2025 Africa.

As a strategic commodity that can be used to tackle food insecurity in West Africa, countries have developed national rice sector development strategies. At the regional level, ECOWAS is playing a key role in supporting member states through regional input and market development policies to address the binding constraints on rice production in the region. West Africa has continued to face a structural rice deficit over the years, spending over USD 4 billion annually on rice imports (Figure 3). FAO has projected that the structural rice deficit will continue to widen in the years ahead, thus making the region increasingly reliant on imports.

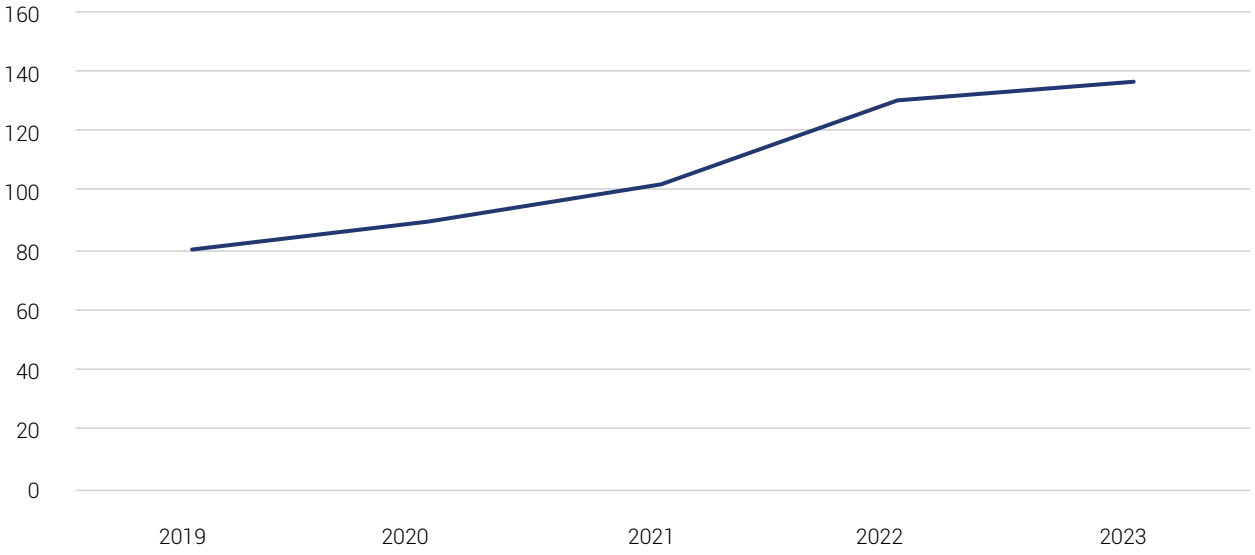
Figure 4 shows the international rice export price index. Prices have been increasing over the past 12 years.

West Africa has continued to face a structural rice deficit over the years, spending over USD 4 billion annually on rice imports (Figure 3). FAO has projected that the structural rice deficit will continue to widen in the years ahead, thus making the region increasingly reliant on imports.



FAO projects that international rice prices will continue to rise due to trade policies of some exporting countries and adverse weather conditions that could impact production in major exporting countries. While the increasing rice price will hurt consumers, it provided opportunities for rice value chain actors in West Africa to increase production, processing, and trade to bridge the supply-demand gap, which is currently filled with imports.

Figure 4: International Rice Export Price Index



Source: Author, based on FAOSTAT

1.3 Women Entrepreneurs in the Rice Value Chain in West Africa

Women and men play important and distinct roles in the rice value chain in West Africa. The roles and responsibilities of men and women in the rice value chain are dynamic and can change over time, depending on the rice-growing ecology. Major driving forces include factors such as shifts from subsistence to commercialized rice production, increasing mechanization, and male emigration. In general, men are mainly involved in land preparation (brushing, ploughing, and levelling) and transportation, while women play important roles in production and postproduction activities. The majority of rice farmers are smallholders who depend on family labor and some hired labor on a seasonal basis. Women are often involved in rice farming activities that are mostly performed manually, such as sowing, transplanting, weeding, harvesting, and post-harvesting activities, including threshing, winnowing, parboiling, drying, and milling.

Women owners of very small, small, and medium enterprises are limited by socioeconomic factors that prevent them from upgrading and expanding their businesses. In the postproduction segment of the rice value chain, women are mainly engaged in lower-skilled activities that are generally manual. This is due to their limited skills and know-how, and their lack of access to the capital needed to purchase equipment, productive resources, and enough rice to supply within the value chain.

Postproduction activities are mostly carried out by women using basic technologies. This is further exacerbated by limited incentives for improving access to postproduction technologies, leading to poor quality milled rice and low prices and profitability of domestic rice. Poor postproduction processes reduce both the quantity and quality of rice available for consumption, therefore undermining all four pillars of food security - availability, access, utilization, and stability. Thus, postproduction management is an important step in ensuring the profitability of rice farming and household and community food security, as well as upgrading WVSEs and WSMEs. In Sierra Leone, and in specific rice-consuming communities in Guinea, Mali and Senegal, where food security is synonymous to rice self-sufficiency, a community that

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is rice self-sufficient is resilient to external shocks and threats. Good postproduction management can bolster that resilience.

While there are several improved rice postproduction machines and equipment available on the market, women face gender barriers to access and adoption. Specifically, women are constrained to adopt certain postproduction technologies due to several factors, such as the design of technology, low incomes, and poor access to credit, land, and other physical resources. Women in rice postproduction activities are also constrained by limited access to information as well as intra-household barriers. Cultural factors at the household level, such as limited participation of women in household decision making and other gender-based role differentiations, affect women's access to agricultural inputs, technologies, and extension services.

Although women dominate the rice retail market, they operate at a microlevel and are confined to roles in the periphery of the rice distribution network, which is dominated by men. This situation makes it difficult for women to access the necessary resources to upgrade their enterprises and to play leading roles in the postproduction segments. Consequently, women are not equally empowered to maximize their generated wealth and connect with key value chain actors to expand their business opportunities.

1.4 Constraints on the Rice Value Chain Process

There are several constraints hindering rice production, widening the gap between supply from the region and the increasing consumption. The ECOWAS Rice fact

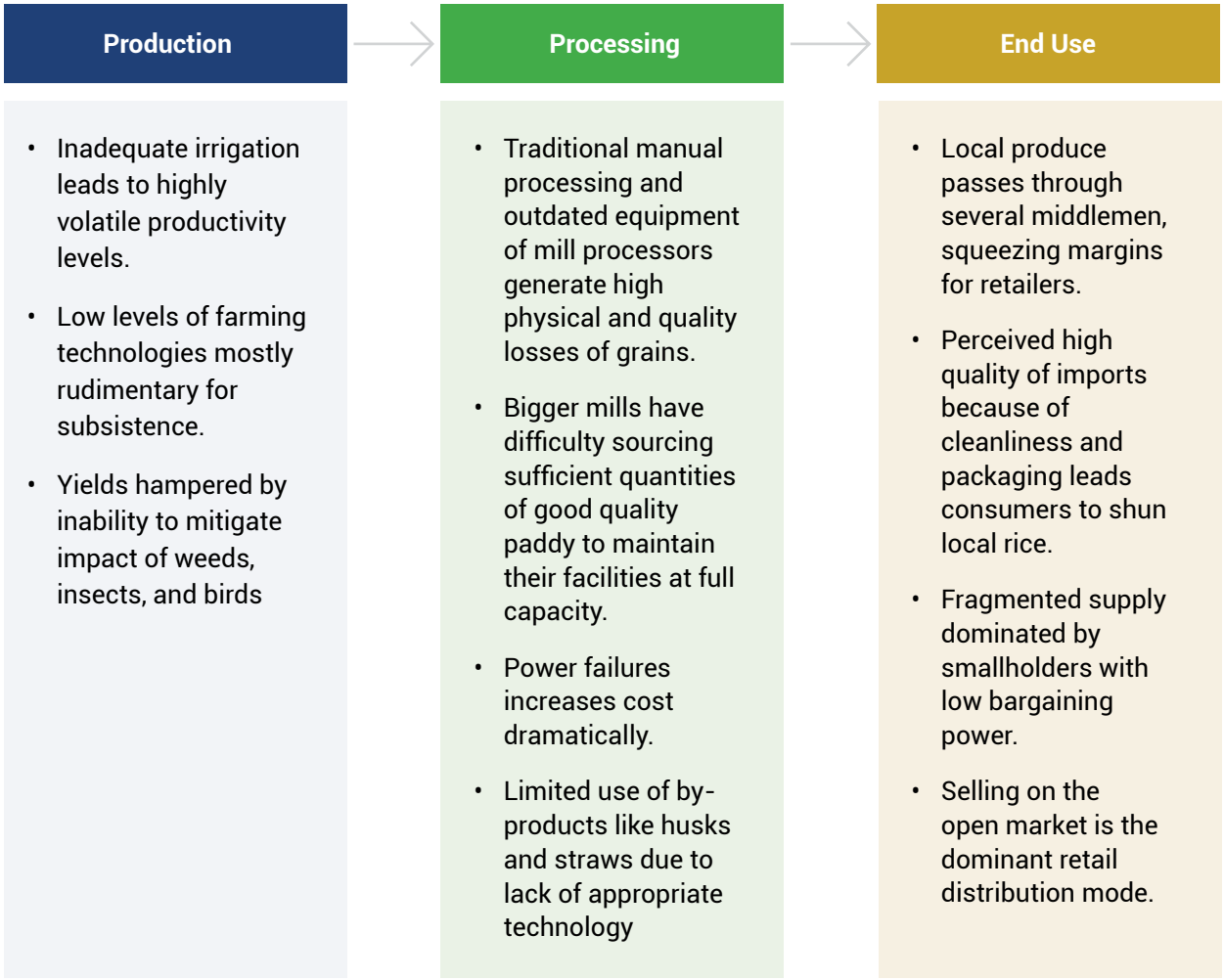


book summarizes the main constraints in production, processing, and end use in West Africa. The key constraints are presented in Figure 5.

Key production constraints include limited irrigated land area, use of low-level technologies, and low productivity. In the area of processing, limited access, use of inefficient technologies, insufficient supply of paddy for higher capacity utilization of mills, poor access to electricity, and limited use of byproducts feature as key bottlenecks. There are limited by-products, leading to lower margins compared to regions where, in addition to milled rice, the sale of byproducts increases the margins of milling companies.

While women play important roles as entrepreneurs at each segment of the rice value chain, the constraints affect them more than their male counterparts, who have access to more resources and options at the household level. As shown by the World Bank, women entrepreneurs earn 34% less profit margin than men entrepreneurs across Sub-Saharan Africa. This is attributed to gender discrimination, lack of a support network, and other social and self-limiting factors (World Bank, 2019)[3].

Figure 5: Key Constraints in the Rice Value Chain Process



Source: ECOWAS Rice Fact Book

3 World Bank Group (2019). World Bank Group. 2019. Profiting from Parity: Unlocking the Potential of Women's Business in Africa. World Bank, Washington.

1.5 The Case for investing in Women Entrepreneurs in West Africa.

Governments and development partners in West Africa have enacted policies and are implementing programs to support WVSEs and WSMEs to succeed. This is in recognition that investing in WVSE and WSMEs can boost economic and social development.. According to the African Development Bank, there is currently a USD 42 billion funding gap for women entrepreneurs in Africa. However, it is projected that Africa could gain USD 316 billion in Gross Domestic Product (GDP) by 2025 if the gender gap is bridged[4]. Thus, supporting women entrepreneurs in any country in Africa is a strategic approach to bolstering the economy. The following paragraphs outline specific justifications for investing in WVSEs and WSMEs in the rice postproduction value chain in West Africa. additional reasons for supporting women entrepreneurs that are applicable to West Africa, based on a review of reports by She.Work[4] and We-Fi.[6].

Boosting economic empowerment and entrepreneurship for Women In West Africa, women dominate the very small and small and medium enterprises segment, and they form a significant group contributing to the economies of West African countries. Thus, by investing in WVSE and WSMEs in the rice postproduction value chain, especially in threshing, parboiling and milling, packaging, marketing and distribution contributes to bolstering local and national economic activities. Such investments will also lead to efficiency along the postproduction rice value chain, thereby contributing to enhance the quantity and quality of rice produced and the supply chain. According to the World Bank,



4 https://www.afdb.org/en/topics-and-sectors/initiatives-partnerships/afawa-affirmative-finance-action-women-africa
5 https://she.work/women-entrepreneurs-challenges-strengths-impact/#:~:text=The%20entrepreneurship%20landscape%20has%20rapidly%20evolved%20over%20the,innovative%20ideas%2C%20and%20a%20fresh%20approach%20to%20business
6 https://she.work/women-entrepreneurs-challenges-strengths-impact/#:~:text=The%20entrepreneurship%20landscape%20has%20rapidly%20evolved%20over%20the,innovative%20ideas%2C%20and%20a%20fresh%20approach%20to%20business



women make up 58% of the self-employed population in Africa, contributing about 13% of the GDP of the continent (WDI, 2024). Therefore, supporting WVSEs and WSMEs in rice postproduction value chain does not only contribute to improving the livelihoods of women. In addition, women consider entrepreneurship as a pathway to economic empowerment, and provides substantial economic benefits for women and economies. Therefore supporting women entrepreneurs in the postproduction rice value chain enables women to accumulate earnings, assets, and economic power that would otherwise have been unattainable, thus reducing economic inequality between men and women.

Expands human capital development. Women are more likely to invest their earnings in their children's health and education, improving their own children's prospects, while expanding the human capital in the respective country. In addition, it is known that women-led businesses tend to employ more women. Therefore, supporting VSEs and WSMEs in rice postproduction value chains, especially through skills development and

> **supporting women entrepreneurs in the postproduction rice value chain enables women to accumulate earnings, assets, and economic power that would otherwise have been unattainable, thus reducing economic inequality between men and women**

capacity building activities required to enhance the technical, managerial skills and competence of women will catalyse job creation for more women, and improve the welfare of their children and families.

Contribution to building resilient communities. Despite gender discrimination in business and other sectors, more WVSEs and WSMEs are more embedded and connected to their local communities. The perseverance of women in the face of entrepreneurship setbacks is the bedrock and quality of their success. Therefore, investing in WVSEs and WSMEs in the rice postproduction value chain contributes to building supportive communities. Specifically, empowered women entrepreneurs more often prioritize and contribute to creating an ecosystem in their community that benefits everyone involved. It is in this spirit that more women entrepreneurs tend to provide mentorship, coaching, and peer support to other women.



Promote innovation, especially to serve women better. Women have valuable insights when it comes to devising technologies and innovations that better serve women. Thus, supporting women entrepreneurs can leapfrog more women to the forefront of technology development that better serves women. This is because women entrepreneurs bring fresh ideas and solutions to the design and development of technologies and innovation, while addressing gaps that might have been overlooked by men in the design stage.

Women entrepreneurs have exceptional adaptability. Women entrepreneurs are known to demonstrate exceptional adaptability in the face of challenges. This quality is essential, especially at a time when economies around the world are facing significant challenges and uncertainties, largely due to supply chain interruptions, rising commodity prices, the pandemic, and regional conflicts. It is under such dynamic and changing business environments that women entrepreneurs have exhibited higher adaptability and resilience.

Women entrepreneurs are known to demonstrate exceptional adaptability in the face of challenges. This quality is essential, especially at a time when economies around the world are facing significant challenges and uncertainties, largely due to supply chain interruptions, rising commodity prices, the pandemic, and regional conflicts.



1.6 Purpose of the Report

The purpose of this report is to provide data and information that could be used to upgrade and advance WVSEs and WSMEs in rice postproduction. The report focuses on the binding constraints that women entrepreneurs in rice postproduction face and identifies applicable solutions to the challenges women entrepreneurs face with respect to access to finance.

The report was prepared for the IsDB under the framework of the EWASME financed from the We-Fi, a financial intermediary fund established to support WVSEs and WSMEs in developing countries. The program aims to stimulate increased support from the public and private sectors to create an enabling environment and open new doors for women entrepreneurs across the developing world, as well as to unlock financing for WVSEs and WSMEs in developing countries.

1.7 Methodology

This report is based on the country reports from Guinea, Niger, Senegal, and Sierra Leone that were prepared as part of IsDB's EWASME program. The preparation of this report involved reviewing and synthesizing the four country reports. Data and information from the country reports were complemented by desk research to enrich the thematic content.

The desk research was conducted in two stages. First, related works on women owned/led SMEs in general, and in West Africa in particular, were reviewed to identify common threads between women owned/led VSEs and SMEs. This was followed by a literature review of the World Bank Enterprise Survey reports and FAO Statistics (FAOSTATS). These provided quantitative data to complement the qualitative and quantitative data from the four IsDB-commissioned country reports.

02

GENDER IN ENTERPRISES IN **WEST AFRICA**



2.1 Women Ownership and Participation in Enterprises

Women in small, medium and large-scale enterprises in West Africa are presented in Table 1, based on the World Bank Enterprise Survey data, which show that, on average, there are fewer women-owned enterprises in manufacturing and related businesses[7] in West Africa, with only 16.4% in manufacturing. Also, the proportion of women top managers in enterprises is limited (about 7%), meaning that the voices and concerns in firms are not advocated for and addressed by women, due to their marginalization in top management. About 16.5% of women are permanent full-time non-production workers, while about 8% are production workers. This data further exemplifies the marginalization of women in formal enterprises.

Table 1: Table Women Ownership and Participation in Enterprises.

Mali	Small	Medium	Large	Average
Percent of firms with female participation in ownership	13.2	13.1	34.0	15.3
Percent of firms with a female top manager	13	8.3	13.6	11.6
Proportion of permanent full-time workers that are female (%)	22.9	14.1	13.6	19.1
Percentage of permanent full-time non-production workers that are female	7.5	16.8	20.1	14.4
Percentage of permanent full-time production workers that are female	7	7.2	14.9	8.3
GUINEA				
Percent of firms with female participation in ownership	7.7	12.6	22.2	15
Percent of firms with a female top manager	6.8	1.7		5.8
Proportion of permanent full-time workers that are female (%)	15.9	10.3	6.9	14.8
Percentage of permanent full-time non-production workers that are female	17.3	21	8.4	N.A.
Percentage of permanent full-time production workers that are female	6.3	4.7	12.1	N.A.
SIERRA LEONE				
Percent of firms with female participation in ownership	19.4	14.6	16.7	18.8
Percent of firms with a female top manager	15.9	16.8	12.1	4
Proportion of permanent full-time workers that are female (%)	23.3	23.7	20	20.8
Percentage of permanent full-time non-production workers that are female	33.4	34.2	27.8	8.5
Percentage of permanent full-time production workers that are female	23.6	24.4	16.8	8.8

Source: Author, compiled from World Bank Enterprise Surveys

7 Excluding agriculture fishing, mining, public utilities, financial intermediation, public administration, education, health, and social work

2.2 Characteristics SMEs and VSMEs

Women entrepreneurs are involved along the value chain activities in various scales: microenterprises (very small), small enterprises, and medium enterprises, together referred to as MSMEs. While the size factor bears significance, countries have varying definitions of MSMEs, and the definitions may also vary across sectors within the same country. However, a common thread across the definitions, countries, and sectors includes size, location, wage labor, target market, and sophistication of inputs. Given this backdrop, the characteristics of MSMEs provided by the International Finance Cooperation (IFC) are adopted for this publication and are summarized in Table 2.

Table 2: Definition of Micro, Small, and Medium Enterprises

IFC MSME Definition				Loan Size Proxy
Indicator	Employees	Total Assets USD	Annual Sales USD	Loan Size at Origination USD
Micro	< 10	<100,000	<100,000	<10,000
Small	10-49	100,000 - < 3 million	100,000 - < 3 million	<100,000
Medium	50-300	3 million -15 million	3 million -15 million	<1 or 2 million

Source: IFC Definition of targeted sectors

IFC has provided a standardized definition of MSMEs. Important variables used to define MSMEs include the value of owned fixed assets, annual sales, and the number of wage-earning employees. In addition to size, the location of these enterprises also matters in the context of West Africa.

Size is the most frequently used variable in defining MSMEs. In describing WMSMEs, IsDB adopts the definition provided by IFC. Therefore, a ***microenterprise*** is a business with a total asset base of less than USD 100,000 and annual sales less than USD 100,000. ***Small enterprises*** have a total asset base of USD 100,000 to less than USD 3 million. ***Medium enterprises*** have a total asset base of USD 3 million to USD 15 million. In the rice postproduction value chain in West Africa, women dominate the small -enterprise category and their ownership or participation decreases as the size of the business increases.

Location is also an important classification variable. In West Africa, microenterprises and small enterprises in the rice postproduction value chain are often operated from homes, on the roadside, or have no fixed address. Medium enterprises may have known addresses, but mostly operate informally.



Labor defines the type of enterprise. Most small and micro-enterprises use family labor or informal apprentices and pay minimal or in-kind wages. In principle, a microenterprise has between 5 and 10 paid workers, while small enterprises have 10 to 49 paid workers, and medium enterprises have between 50 and 300 workers.

Produce for domestic market: Micro, small, and medium enterprises in West Africa produce goods and services that target the domestic market, which are subject to high competition and volatile demand. In some cases, they create backward and forward linkages within the country's economy by using locally produced inputs as well as producing inputs for larger industries. In terms of technological sophistication, they use labor intensive technologies that are less sophisticated and costly than their larger counterparts. The characteristics of micro, small, and medium sized enterprises and their implications for women are listed in Table 3.

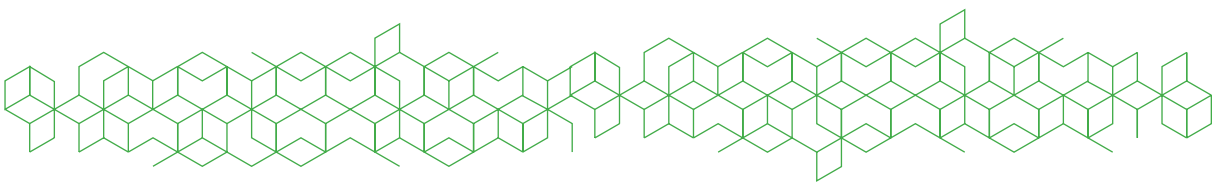
Table 3: Characteristics of VSE and SMEs and Implications for Women

	Structure	Constraints	Skills/ Knowledge	Location	Hiring Practices
Characteristics of very small, small, and medium enterprises	- Flexible - Loosely structured - Not present in all sectors - Underperforming	- Less education, training, and experience are required - Lower level of capitalization required - Use of family labor rather than wage labor	- Can use already existing skills - Limited skills - Acquire skills by doing	- Street location - Homebased	Women dominate as owners of certain businesses
Implications for women	Enable women to divide their time between domestic responsibilities and economically productive activities	Because there are few barriers to entry into the micro and small enterprise sector, women can easily participate	Traditional activities such as garment making and food processing can be turned into businesses	Women can be involved in several activities at a time and overhead cost is low	Women entrepreneurs tend to employ women employees

The majority of business enterprises in West Africa are in the very-small, small, and medium size categories. They constitute an important component of the private sector. It is estimated that about 95% of total enterprises fall in this category, and are a major source of income and employment in the sub-region. These enterprises can be found in almost all productive and service sectors. They have flexible structures compared to the large-scale enterprises and firms. They tend to be informal and have little or no barrier to start up. This allows women to combine domestic responsibilities and economically productive activities. In general, very small, small, and medium enterprises face several constraints that have implications for their long-term survival. Thus, they face high rates of failure, especially during the first five years of start-up.

African SMEs face major challenges. Top on the list are: (i) lack of access to finance; (ii) poor access to electricity supply; (iii) poor management; (iv) lack of competency and

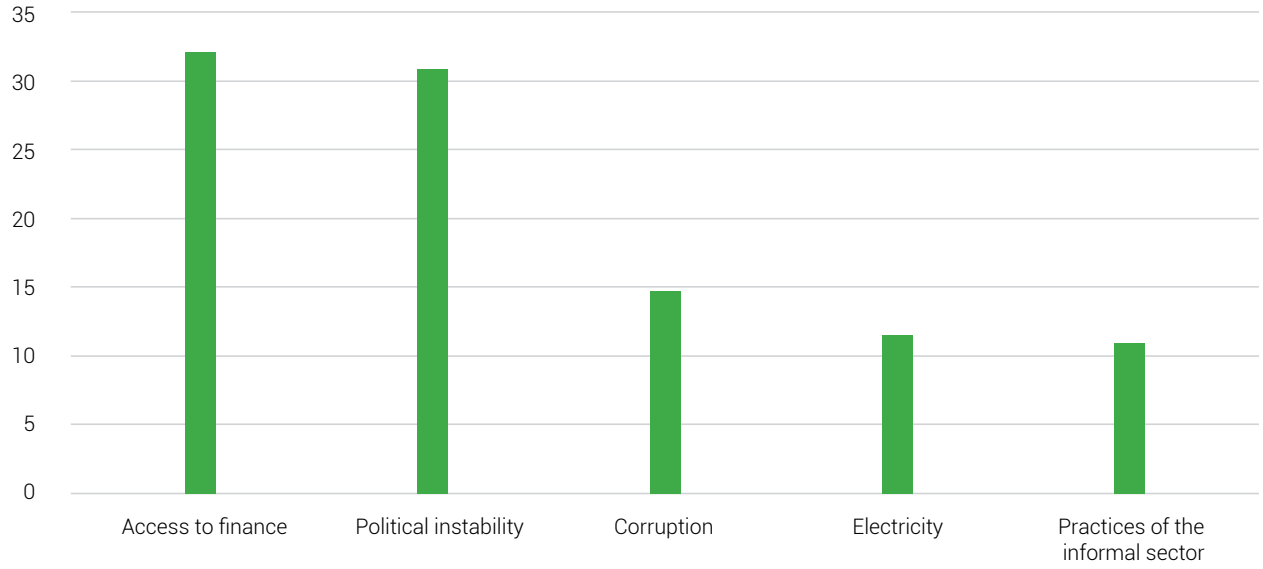
Most small and micro-enterprises use family labor or informal apprentices and pay minimal or in-kind wages. In principle, a microenterprise has between 5 and 10 paid workers, while small enterprises have 10 to 49 paid workers, and medium enterprises have between 50 and 300 workers.





capability; (v) negative perceptions; (vi) lack of access to reliable information; (vii) limited government support; and (viii) corruption. Other challenges include political instability, labor issues due to the lack of established legal frameworks, lack of coordination, ethnic violence, destruction, and lack of qualified personnel (Bowen et al., 2009; Katua, 2014). A further review of these constraints specific to West Africa, using the World Bank Enterprise Survey data, identified the five largest obstacles to businesses in West Africa in order of significance as follows: (i) lack of access to finance; (ii) political instability; (iii) corruption; (iv) poor access to electricity; and (v) practices of the informal sector (Figure 6).

Figure 6: Largest Obstacles to Businesses in West Africa

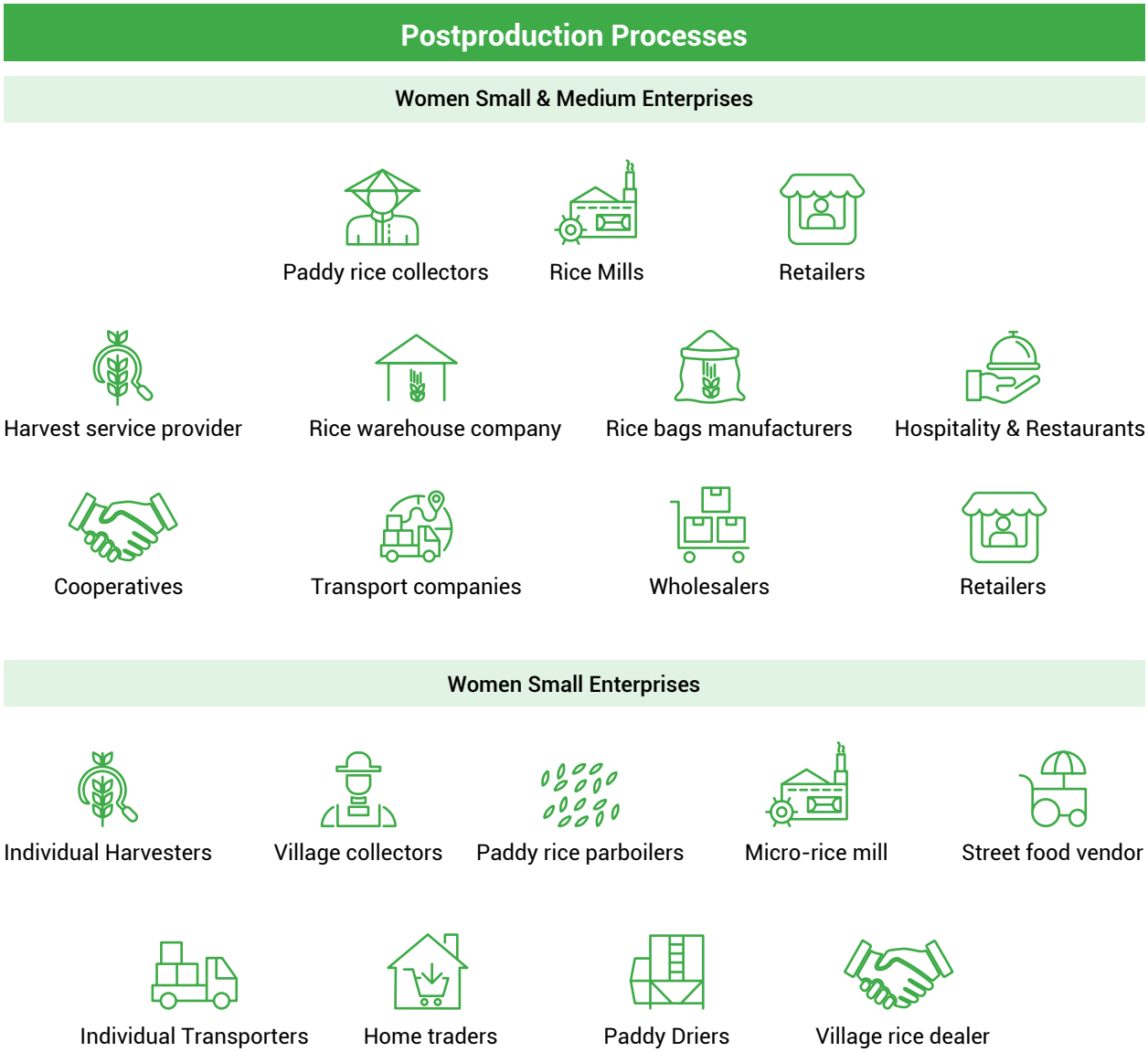


Source: Author, based on World Bank Enterprise Survey Data

2.3 Women in Postproduction Segments of the Rice Value Chain

Figure 7 provides a taxonomy of WVSEs and WMSEs in rice postproduction. WVSEs are dominant in the lowest stages of the postproduction processes, with individual business ownership structures. Though formal, decisions are largely made by the proprietors, and such businesses are much riskier than WSMEs, which tend to be more formalized with a board that meets regularly to review and approve investment plans. WSMEs are more prominent in the postproduction process transition towards higher and modern technologies and markets. It is at this stage that WSMEs could play a bigger role in connecting different components along the rice value chain. In addition, as rice postproduction processes, especially milling, become more commercialized and industrialized, SMEs play greater roles in linking other segments of the value chain from production and processing to market. However, the prominence of women ownership declines as businesses become larger, more formalized, and use more sophisticated technologies.

Figure 7: Taxonomy of WVSEs and WSMEs



Source: Author.



Postproduction Processes and Technologies

The main postproduction processes from harvesting to marketing (wholesale and retail) and the technologies used are shown in Figure 8.

Figure 8: Postproduction Processes and Technologies

Process						
Harvesting	Threshing	Parboining	Drying	Storage	Milling/ Processing	Marketing Wholesale/ Retail
Process						
• **Traditional** - Sickles - Knives • **Mechanized** - Manual Ripper • **Combined** - Harvester - Wagon/sacs	• **Traditional** - Sticks - Treadle threshers - Trampling • **Mechanized** - Machine threshers - Mobile - Stationery	• **Traditional** - Sun - Mats - Pavement • **Mechanized** - Heat blower . Batch . Flow . Solar	• **Traditional** - Sun - Mats - Pavement • **Mechanized** - Heat blower . Batch . Flow . Solar	• **Traditional** - Storage - Home - Bags - Jute - Woven plastic	• **Traditional** - Mortar & pestle • **Mechanized** - Micro-mill - Two stage mill - Modern rice mill - Miller & Destoner Polisher/ Whitener - Grader, Scale - Bagging machine	• **Traditional** - Cup, Bowls, Tin (various sizes) - Scale - Bags - **Shed** - **Store** - **Truck** - **Supermarket**

Figure 9: Rice Postproduction Activities and Losses (percent)

Gathering and Field/ shed Storage
0.5-1%

Transport to pre-threshing Location
0.5-1.2%

Threshing, Winnowing and Bagging
2-5%

Transport to Storage
0.5%

Storage
2-5%

Drying and Parboiling
1%

Milling and Bagging
3-6%

Storage and Marketing
1%

Source: Author, based on field interviews in Sierra Leone 2021

2.4 Postproduction losses

A schematic presentation of rice postproduction activities and associated losses among WVSEs and WSMEs is presented in Figure 9. Cumulatively, WVSEs and WSMEs lose approximately 10.5% to 22% of the rice during the postproduction phase, thus negatively affecting incomes. These losses are associated with the financial and non-financial constraints faced by WVSEs and WSMEs.

As a consequence of limited financial capacity, WVSEs and WSMEs largely use manual labor (mostly women) and rudimentary machines and equipment to perform postproduction activities, thus leading to high postproduction losses. The drudgery of these manual activities affects the health of women, especially in rural areas with limited and poor access to health facilities.



03

FINANCING WVSEs AND WSMEs IN RICE **POSTPRODUCTION**



3.1 Barriers Facing Women Entrepreneurs

WVSEs and WMSEs in West Africa face multidimensional problems compared to their male counterparts, mainly due to socioeconomic and cultural barriers. In particular, women have major responsibilities, having to care for family and children, which places them before more obstacles than men for establishing and managing small or medium enterprises on a fulltime basis. Relatedly, they are naturally disadvantaged in raising the initial capital, as they have limited personal assets that can be utilized to raise funds to start or grow a business. Information on financial and investment opportunities is mostly more available to men than women, thus women are frequently crowded out of opportunities by men. Women are also constrained to have business relationships with different financial institutions due to gender-related cultural barriers.

In addition to the above constraints, WVSEs and WMSEs are severely constrained in terms of accessing financial services. This is associated with the high-risk profile perceived by financial institutions, smaller size, informality, limited experience, and undocumented performance, as well as the sector in which they usually operate.

Table 4 shows the various sources used to finance purchases of fixed assets (investments). The percentage of businesses with a checking or savings account and the percentage of businesses with a bank loan are two indicators that show the use of financial services by private firms. While the percentage of firms with a checking or savings account measures the use of deposit mobilization services, which helps firms manage their liquidity and payments, the percentage of firms with a bank loan measures the use of financial services on the credit side.

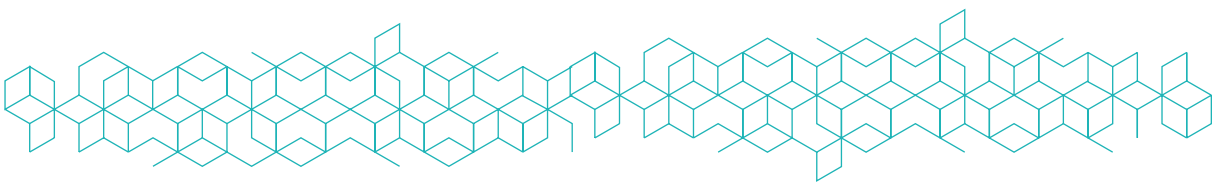
Table 4: Sources of Financing for Purchase of Fixed Assets

	Countries			
	Guinea	**Mali**	**Senegal**	**Sierra Leone**
Percent of firms with a checking or savings account	98	95	78	70
Percent of firms with a bank loan/line of credit (SMEs)	2	14	23	9
Percent of investment financed internally	92	75	72	91
Percent of investment financed by banks	3	19	7	1.3
Percent of investment financed by supplier credit	0	0	7	3.3
Percent of investment financed by equity or stock	4	2	4	3.8

Source: Author, Compiled from World Bank Enterprise Surveys Country Reports

Using data from World Bank Enterprise Survey[8] reports and complemented by information from case studies from Guinea, Mali, Senegal, and Sierra Leone, external sources of finance to enterprises in general can be derived. Most importantly, WVSEs and WSMEs are constrained by the lack of external financing for both working capital and purchase of fixed assets. In general, the Enterprise Survey classifies the different sources of external finance for purchase of fixed assets into four categories – equity finance and three options for debt finance, which is further subcategorized into formal debt finance (including banking and non-banking financial institutions), trade finance (includes credit from suppliers and/or customers), and the Other category, which includes informal sources of credit, such as moneylenders, friends, relatives, etc.
Ownership of a bank account by a WVSE or WSME is one indicator of formality. While over 70% of the enterprises owned a bank account, those that had access to formal bank loans vary, at 2% in Guinea, 9% in Sierra Leone, 14% in Mali, and 23% in Senegal. From the table, it is clear

8 World Bank Enterprise Surveys – www.enetrprisesurveys.org

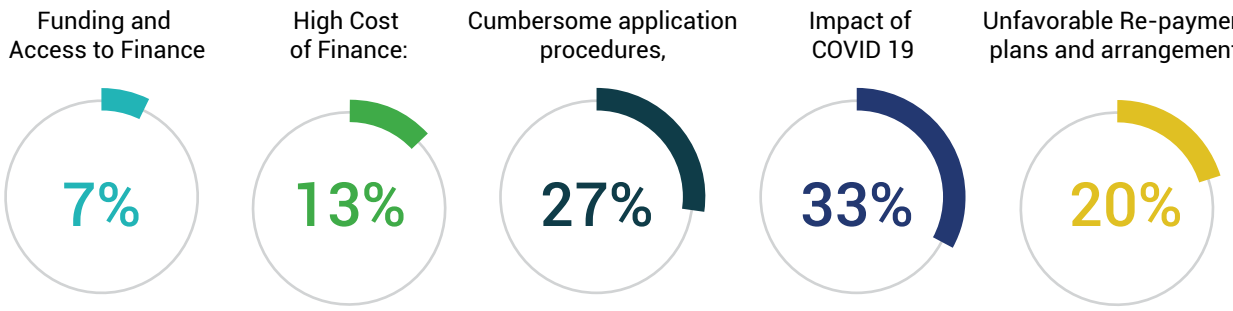




that SMEs consistently rely more on trade credit and informal sources to finance the purchase of fixed assets. It is also noteworthy that enterprises in the West Africa region rely mostly on the use of informal credit to finance investments in fixed assets. The use of formal debt to finance investments is low among SMEs, due to financial and non-financial constraints that hinder access. As with the financing of fixed assets, the use of formal credit sources by SMEs to finance working capital is equally low.

Agricultural finance is one of the most difficult types to secure. African women farmers experience greater constraints than their male counterparts. In the rice postproduction value chain, this constraint must be understood within the context of the business-financing environment. In general, agribusiness financing in West Africa is still in the developmental stage[9] and is inefficient, as it does not provide the necessary services to private businesses, especially those owned or led by women. Banking products are not segmented by enterprise, size, and needs of WVSEs and WSMEs, nor do they facilitate funding for the purchase of fixed assets. In order to contextualize the financial constraints specific to WSMEs, IsDB commissioned a study in Guinea, Mali, Senegal and Sierra Leone, whereby women entrepreneurs were asked in a group interview to rank the main constraints affecting their businesses. Results from the group interviews are presented in Figure 10.

Figure 10: Ranking of Financial Constraints on WVSEs and WSMEs in West Africa



Source: Author, Sierra Leone

9 A well-developed financial market does not only provide payment services, mobilize deposits, and facilitate funding for the purchase of fixed assets, but also provides support to businesses by connecting firms that are creditworthy to a broad range of lenders and investors.

3.2 Financial and Non-Financial Constraints on WVSEs and WSMEs

3.2.1 Financial Constraints

1. Limited Access to Finance: Access to finance is the most significant constraint on the growth and transition of WVSEs and WSMEs to the next level. Lack of access to finance diminishes the opportunity of WVSEs and WSMEs undertaking rice postharvest activities to acquire the technologies they need to improve competitiveness, rice quality, and access to market. There are several factors that hinder WVSE and WSME access to finance. WVSEs and WSMEs undertaking rice postproduction activities in West Africa operate in a generally risky economic environment. This situation is compounded by the perception banks and other financial institutions have of WVSEs and WSMEs as high-risk customers, as they lack acceptable collaterals and assets. In addition, WVSEs and WSMEs active in rice postproduction are constrained to access loans due to property rights and legal regulations that often restrict women's access to and control over assets that can be acceptable as collateral, such as land, houses, and other household assets. The lack of awareness of the financing options and products available in banks and microfinance institutions, as well as the biased attitude of banks and lending institutions, cultural norms, and family responsibilities, affect the access of WVSEs and WSMEs active in rice postproduction to finance.

Lack of access to finance diminishes the opportunity of WVSEs and WSMEs undertaking rice postharvest activities to acquire the technologies they need to improve competitiveness, rice quality, and access to market.

2. High Cost of Finance: The generally high cost of credit offered by banks and other financial institutions, which correlates to poor access to finance, is the second most significant financial constraint on WVSEs and WSMEs. Commercial-bank lending to WVSEs and WSMEs active in rice postproduction is not common across West African countries, as they are perceived as non-resilient with poor management structures and without the necessary knowledge and skills to develop business plans and other documents required by financing institutions. Thus, where finance is made available, mostly by microfinance institutions, the interest rate can be as high as 300% per annum. In addition, the repayment plans are unfavorable and do not align with the cash flows of the WVSEs and WSMEs. Consequently, the few WVSE and WSMEs in the rice postproduction value chain that have accessed formal credit consistently had difficulty making regular repayment, thus leading to a high-default rate.

3. Unfavorable Repayment Plans: Repayment plans are unfavorable and do not align with the cash flows of the WVSEs and WSMEs. Consequently, the few WVSE and WSMEs in the rice postproduction value chain that have accessed formal credit consistently had difficulty making regular repayment, thus leading to a high-default rate. These reasons include (a) seasonal nature of agribusinesses due to the inherent seasonal nature of farming cycles that are influenced by the climatic condition.

4. Exclusion from Financial and Banking Services: This is the third most significant financing constraint facing WVSEs and WSMEs in the rice postproduction value chain in West Africa, where they are excluded on grounds related to size, physical location, and distance from credit source. Although there are financial products that can be accessed online, the cost of internet access and limited access to electricity, especially in rural areas, hinder the adoption of such products. As the majority of VSMEs and WSMEs active in rice postproduction are informal, they lack the necessary identification and certificates to access loans from formal financial institutions.

5. Cumbersome Application Procedures: This was ranked as the fourth most significant constraint affecting WVSEs and WSMEs in the rice postproduction value chain. In particular, banks require VSMEs and WSMEs to open accounts in order to access loans. However, data from a survey in Sierra Leone showed that although most VSMEs and WSMEs did open bank accounts, less than 10% of them acquired loans. With respect to application procedures, VSMEs and WSMEs are required to provide, in addition to business registration documents, official

identification documents (passport, national ID card, voter ID card, driver's license, etc.) of the managers. Acquiring such documents poses an additional difficulty for WVSEs and WSMEs, who may not have the recognizable street addresses and public utility bills required to acquire national ID cards. Thus, the application procedures further affect access to finance for WVSEs and WSMEs. Poor financial literacy amongst WVSEs and WSMEs also contributes to the poor understanding of application procedures, repayment terms, and other loan conditions.

3.2.2 Non-Financial Constraints

There are also non-financial constraints that limit the access of WVSEs and WMSEs to finance. Table 6 presents the ranking of non-financial constraints.

Poor access to rice mills: Access to good quality milling is the most significant non-financial constraint affecting WVSEs and WSMEs in the rice postproduction value chain. When asked which of the two groups of entrepreneurs are most affected, the consensus among the group was that WVSEs are most affected by poor access to rice mills, due to low volumes and long transport distances, which ultimately increase cost and reduce profit margins. While access to rice mills is important, access to good quality modern rice and good rice variety is even more important. Theoretically, the rice milling recovery rate is between 71% and 73%. However, in practical terms, it is only possible to obtain between 68% and 70% from a good variety of paddy. It was estimated that about 3% to 6% of harvested rice is lost through milling and bagging activities.

There are broadly two categories of rice-milling techniques

Table 5: Ranking of Non-Financial Constraints on WVSEs and WSMEs

Non-Financial Constraint	Rank
Poor access to rice mills	1
Absence of paddy drying Facilities	2
Threshing and winnowing	3
Poor storage facilities and access	4
Poor access to energy and water	5
Shortage of labor for postproduction activities	6

Source: Sierra Leone Country Report



in West Africa; manual and automated. WVSEs mostly rely on manually-operated pounder or mortar and pestle to mill rice in villages. The lack of an assured source of energy to run automated rice mills leaves WVSEs with the only option of employing family members, mostly women and children, to mill rice using mortar and pestle. Automated rice-milling machines range from generator-powered single-pass rice mills to sophisticated modern rice mills. According to FAO, milling losses can be reduced by adopting small-scale modern rubber roll shellers and introducing parboiling of paddy before milling.

Absence or low quality of rice-drying facilities: This was ranked as the second most significant non-financial constraint faced by WVSEs and WSMEs in the rice postproduction value chain. It was estimated that 1% of harvested rice is lost as a result of the absence or low quality of rice-drying facilities. The unavailability of a good rice drying facility also increases loss due to poor quality, which further affects the safe storage and transportation and quality of processed rice. The common method of drying paddy is by spreading it on specially prepared concrete drying floors, on trampolines, on mats, or on the sides of tarmac roads. Improved automated methods and equipment have been developed for drying paddy, including solar driers and flash driers. Among the paddy-drying options available to WVSEs and WSMEs in the rice postproduction value chain, sun-drying remains the cheapest in Sierra Leone. Although automated paddy

driers can be purchased from the external market or manufactured locally, the chance of adoption by WVSEs and WSMEs is limited due to several factors, including: (i) high cost of fuel to run the equipment; (ii) the small quantities of paddy available to WVSEs and WSMEs can be more economically dried in the sun; (iii) lack of knowledge about such equipment; and (iv) the high capital outlay to purchase such equipment.

Unavailability of labor-saving threshing equipment: Difficulties in threshing were ranked the third most significant non-financial constraint affecting WVSEs and WSMEs in the rice postproduction value chain. The manual threshing method is the most common in Sierra Leone. This involves trampling by women on the panicles, beating the panicles with sticks, or hitting the panicles on wood. Trampling leads to rashes in the feet and beating with sticks results in swollen patches in the palm. Threshing is commonly done on bare floors, concrete floors, and/or the sides of tarmac roads. All of these increase impurities; stones and other foreign materials, in the threshing operation, thereby leading to loss in quality and market value of the milled rice. Winnowing is also a manual operation exclusively done by women. Threshing and winnowing are tedious and time-consuming operations. With the increasing shortage of farm labor in villages, the cost of threshing and winnowing is increasing and, thus, contributes to lowering the margins received by WVSEs and WSMEs in the rice postproduction value chain.

Mechanized small-scale threshing and winnowing equipment can be purchased or manufactured locally. However, the lack of knowledge and the capital outlay to purchase such equipment is a challenge for WVSEs and WSMEs in the rice postproduction value chain in Sierra Leone. Poor access to efficient threshing equipment contributes to the loss of 2% to 5% of the total harvest, which, when combined with the cost of labor for threshing, further depresses the margin of WVSEs and WSMEs in the postproduction rice value chain.

Poor storage facilities: Poor storage and access to storage facilities for paddy was ranked as the fourth major non-financial constraint affecting WVSEs and WSMEs in the rice postproduction value chain. Rice storage occurs at the farm following harvest and in the village, awaiting threshing. It is therefore necessary that proper storage infrastructure be provided at each stage for better handling and processing of paddy. However, this is not the case for WVSEs and WSMEs in the rice postproduction value chain, as paddy is stored in traditional structures or at homes. It was estimated that improper storage can affect as much as 1% of the harvested paddy.

Lack of knowledge and capacity to use post-harvest technologies and equipment: Most WVSEs and WSMEs in the rice postproduction value chain have poor knowledge, know-how, and access to modern technology and information for rice postproduction and for making managerial decisions. Therefore, it is crucial to improve the knowledge of women in modern and improved postproduction technologies and the use of relevant equipment. Accordingly, targeted training for WVSEs and WSMEs is required, as emerging rice-farming and postproduction operations are becoming more knowledge-intensive. WVSEs and WSME could also be supported to participate in exchange visits within the country to see first-hand how other successful micro, small, and medium enterprises in the rice postproduction value chain are operating and being managed.

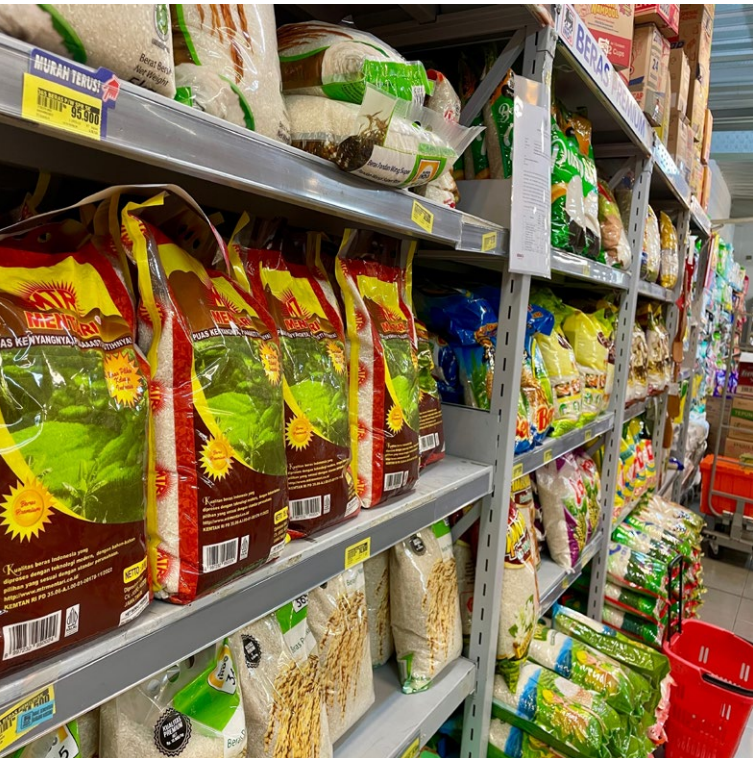
Shortage of labor for rice postproduction activities: Shortage of labor cuts across all postproduction activities for women entrepreneurs. There are both economic and social factors associated with the shortage of labor in rural areas, where most of these entrepreneurs operate, that are not under the control of the women. This further justifies the need for the introduction of labor-saving technologies and mechanization of rice postproduction value-addition. An option to improving rice postproduction activities and economically empowering women should involve the promotion of labor-saving technologies and mechanization of postproduction activities. Mechanization increases labor productivity and reduces the drudgery involved in various postproduction operations, such as harvesting, transporting, threshing, drying, and milling.

Poor access to efficient threshing equipment contributes to the loss of 2% to 5% of the total harvest, which, when combined with the cost of labor for threshing, further depresses the margin of WVSEs and WSMEs in the postproduction rice value chain.

Poor capacity of financial institutions: In addition to the demand-side constraints mentioned above, the limited capacity of formal financing institutions exacerbates the poor access of WVSEs and WSMEs to finance. A review of the formal financing institutions reveals that the banks have weak capacities to develop profitable and broadened financial products to address the financial needs of WVSEs and WSMEs. Despite the importance accorded to agriculture and agribusiness in the development of West African countries, the majority of banks reviewed do not have an agricultural lending department or unit. Thus, there is a high disparity between SME bank loans for trade (more than 90% of SME loans) as compared to less than 10% for agriculture and agribusiness loans.

3.3 Importance of Access to Finance for WVSEs and WMSEs

Women entrepreneurs constitute the largest number of entrepreneurs in West Africa. Financing women entrepreneurs in any country contributes to a significant boost of the economy. As role models to their families and society, financing women entrepreneurs is a direct contribution to the economic empowerment of women entrepreneurs, through which jobs created directly increase the number of women in employment, who are presently marginalized in formal productive sectors.



Studies across the world by the World Bank, the European Investment Bank (2019, 2021, 2022), UNIDO and others have shown that women-led firms are more likely to have sound management practices in place. Such firms adopt mechanisms for setting smart performance indicators and are more willing to introduce new products and innovative solutions. This may also support gender diversity in the workplace, which is known to usher in more efficient solutions and better decision making. Supporting women-led enterprises bridges the existing gender gap and, thus, contributes to the universal goal of achieving gender parity, enabling women to leverage their potential in the transformation of the economy and society at large. Women-led firms generate further positive spillovers, with their greater use of training courses (40% of women-led firms compared to 34% of male-led firms), thus contributing to developing the human capital of the firm and empowering employees even beyond. Women also attract and retain more women, positively affecting female employment and reducing gender disparities, while promoting more inclusive growth. Notably, 47% of women-owned firms have more than 50% female workers, while 74% of men-owned firms have less than 50% female employees (EIB, 2022).

Women also tend to support the green transition more, attentively monitor CO_2 emissions, and set energy-efficiency targets. Women-led companies have higher environmental, social, and governance scores, which improves institutional evaluation and perception by investors and clients. Gannon, K. E., et. al. (2022) identified gendered barriers and enablers to private sector adaptation, suggesting that women entrepreneurs face a "triple differential vulnerability" to climate change, wherein they: (i) are often more sensitive to climate risk, as a result of their concentration on certain sectors and types of enterprises (e.g., MSMEs in the agricultural sector in remote regions); (ii) face additional barriers to adaptation in the business environment, including access to finance, technologies (climate and adaptation), information, and supportive policies; and (iii) are often concurrently at the frontline of managing climate risk at household levels. Thus, financing women entrepreneurs will contribute to simultaneously alleviating the triple differential vulnerability to climate change, hence enhancing positive climate change actions.

Although both men and women entrepreneurs face skills gaps, which is one of the biggest obstacles in business, women are more strongly affected by this, due to their other household commitments and social factors. These

factors inhibit women-led enterprises from accessing skill-training programs on an equal footage with their male counterparts. Facilitating access by women-led enterprises to skills training programs contributes not only to improving the business outcomes, especially increasing profits, but also to human capital development.

3.4. Providing Value Chain Finance to Women-led Enterprises

Value-chain financing allows enterprises and firms to operate and transact with others and to upgrade. It involves providing financial services and products to one or more points in a value chain to reduce the risk and cost of financing and increase the efficiency and competitiveness of the value chain. Figure 11 illustrates the rice postproduction value chain financing needs of WVSEs and WSMs in Sierra Leone. While this may not be representative of the rest of West Africa, it provides an illustration of the investment needs across the value chain. As the financing needs of WVSEs and WSMs differ across the postproduction value chain, country-specific investment needs must be ascertained through a country-level value chain analysis[10]. Therefore, the appropriateness of postproduction financial products depends on the segment of the value chain that is targeted for upgrading.

It is clear that the estimated financial need of WVSEs and WSMEs for each postproduction process in Sierra Leone depends on the level and sophistication of technology. From the postproduction rice chain analysis in Sierra Leone, the investment needed for harvesting and transfer ranges from USD 400 to USD 1,000 per enterprise. For threshing and winnowing, an investment outlay of USD 400 to USD 3,000 is required per enterprise. For aggregation and storage, the estimated financial need ranges between USD 800 and USD 46,000[11] per enterprise. A postproduction women-led enterprise would require USD 2,500 to USD 10,000 to purchase and install a modern and energy-efficient parboiling facility and construct a drying floor, and about USD 4,500 to purchase and install a modern rice mill (2-2.5t/hr.). For the construction of a modern retail store and outlet, a woman entrepreneur in rice postharvest would need about USD 45,000 to USD 50,000.

Figure 11: Financing Needs of WVSEs and WSMs Along the Postproduction Value Chain

Functions	Harvesting and Transfer	Threshing and Winnowing	Aggregation and storage	Parboiling / Drying	Milling	Storage and Sales	Consumption
Main Actors	• Farmers • Collectors	• Farmers • Collectors	• Transporters • WVSE/WSMEs	• WVSE/WSMEs • Farmers	• WVSE/WSMEs • Millers	• WVSE/WSMEs • Retailers	Consumers
Activities	Gathering and transfer to thresing site	Manual/ mechanical threshing & winnowing	Transfer to storage facilities	Parboiling and drying	Milling, destoning, sorting grading, weighing and bagging	Storage, wholesale and retailing	Consumption
WVSE/WSME Financing Needs	• Harvesters • Bags • Transport	• Threshing Machine • Winnower	• Transport • Storage Facilities	• Drying Floor • Parboiling Facility	• Milling Machine • Destoner machine • Bagging machine	• Store • Weighing Machine	Quality and Competitiveness
Priority	Low	Low	Moderate	High	High	High	Quality and profitability
Loan duration and purpose	Buy harvesting equipment, bags and pay transport cost	Pay user fees or loan to buy thresher and winnower	Construct storage facility/pay transport or buy	Construct drying floor and buy parboiling machine/ equipment	Buy milling machine destoner and bagging machine	Construct store and facilities	Quality and Competitiveness
Amount USD (est.)	400 - 1,000	400 - 3,000	800 - 46,000	2,500 - 10,000	500 - 120,000	45,000 -50,000	Competitiveness and Growth

Source: Author

10 UNIDO, Pro-poor value chain development: https://www.unido.org/sites/default/files/2011-12/Pro-poor_value_chain_development_2011_0.pdf
11 USD 46,000 includes construction of and equipping a standard rice storage facility

3.5 Postproduction Value Chain Financing Types

The most critical points for financing along the rice postproduction value chain include aggregation and storage, parboiling, drying and milling, and sales. The value increases along the chain as the paddy moves from threshing, drying and milling, to sales. However, it must be noted that the quality at each segment of rice postproduction value chain should be assured to gain consumer acceptance and, thus, the market. This is because there is stiff competition between imported rice (preferred by consumers) and domestic rice, due to quality differences.

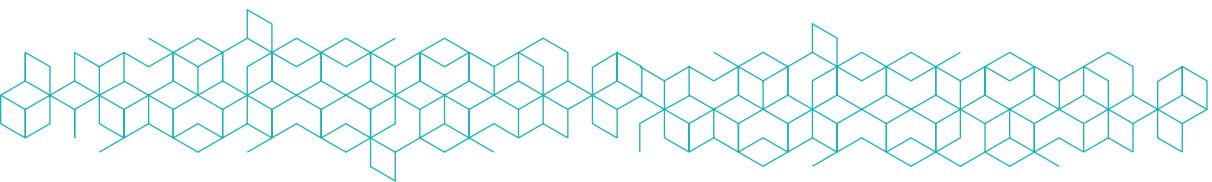
The type of value-chain financing for the various segments of the rice postproduction value chain for WVSEs and WSMEs are illustrated in Table 7. Financial needs at each segment of the rice postproduction value chain depend on the process and on the sophistication of technology used or owned. The more advanced and mechanized the process, the more financial outlay is required to own it. Most WVSEs pay for services, while WSMEs tend to aspire to purchase machines and equipment, with the purpose of upgrading or growing the business.

The decision to own and operate or hire any postproduction machine or equipment by WVSEs and WSMEs is guided by the totality of the operations, not just the purchase price

Table 6:Value-chain Financing Needs of WVSEs and WSMEs and Types of Finance

Rice Value Chain Segment	Need/Purpose	Type of Finance
Harvesting and threshing	Working capital to pay for harvesting and threshing of paddy; purchase simple tools	• Overdraft • Term loan
	Fixed assets (purchase mechanized thresher with winnower and accessories)	• Revolving credit line • Asset-based finance – factoring (accounts receivable), inventories etc.
Transport	Working capital to pay for transport of goods (paddy)	• Overdraft • Term loan
	Fixed assets (purchase truck for paddy and rice transport)	• Vehicle & asset finance
Processing	Working capital (including advance payments to suppliers)	• Overdraft • Revolving credit line • Asset-based finance (accounts receivable), inventories, etc.
	Fixed assets (plant, property)	• Asset finance (leasing, rental, etc.) • Commercial property finance (warehouses)
	Equipment (machinery, capital equipment)	• Term loan • Vehicle & asset finance (leasing, rental)
Storage	Working capital to pay for storage space	• Overdraft • Term loan
	Fixed Assets (construct store)	• Revolving credit line • Asset-based finance
Wholesale, retail, & marketing	Working capital	• Overdraft • Revolving credit line
	Fixed assets (incl. wholesale warehouses, transport vehicles, etc.)	• Term loan to manage cash flow • Commercial property finance • Vehicle & asset finance

Source: Author



of the machine or equipment and other direct costs. The issues to be considered by WVSEs and WSMEs to purchase and own or "pay and use" (contract hiring), and by financing institutions for the provision of any kind of financial product, are listed in Table 8.

Table 7: Decision to Own or Hire Farm Machines and Equipment

Decision to own /Hire Depends on	Ownership	Contract hiring
Timeliness of operation	Machine is ready when needed	Contractor may not be available when needed
Repair, maintenance	Own repair and maintenance needed to reduce downtime	No expertise needed for repair, maintenance
Quality of work	Quality of operation is under control of farmer	Quality of operation is under control of contractor
Cost	Can be lower if there is enough paddy	Costs not dependent on quantity of paddy
Skills required for operation	Operational skills required	No skills required at the farm
Finance requirements	Capital outlay and financing needed	No large capital outlay needed

Source: Author



04

RECOMMENDATIONS ON ADDRESSING THE **CHALLENGES FACED BY WVSEs AND WSMEs**



4.1 Alleviate Financial Constraints

Following the IFC definition of small and medium scale enterprises and women owned enterprises, it is to be noted that that WVSEs and WSMEs have different scales of capital base. As such, the recommendations should be applied within the context of women in each category (very small, small, and medium enterprises). Therefore, organizing WVSEs into manageable groups of 10 to 15 members and building their capacity specifically on accessing and managing group loans, bookkeeping, and basic business management is a desirable condition to reduce their associated risks and loan-management costs. Organizing WVSEs can be financed by development projects and by the national agency responsible for SMEs in each country. Therefore, it is recommended that lending to WVSEs in the postproduction segments of the rice value chain be done in the form of group-credit after organizing, training, and registering the group as a legal entity. A similar process could be applied to the small enterprise category.

4.1.1 Provide Innovative Products for Women SMEs and VSMEs

Accessibility (the ability to access finance) and affordability (the cost of capital, or how much it costs for an enterprise to borrow or receive an investment) constitute the most significant challenges facing VSMEs and WSMEs in West Africa. A major reason is that SMEs are perceived as high-risk clients. For WSMEs that are on the growth trajectory, blended finance provides an opportunity to access the capital required to grow. It is a de-risking instrument that allows WSMEs access to finance. Most blended finance continues to come from philanthropy and bilateral donors using their grant funding. This means that there is potential to expand blended finance programs through the significant resources available to development financial institutions (DFIs), although many do not have concessional windows.

Blended Finance for Reducing Cost of Finance: Blended finance refers to the strategic use of development finance and philanthropic funds to mobilize private capital for sustainable development[12]. This approach is particularly relevant for women-led small and medium enterprises (WSMEs) in agribusiness[13], as it can help address various financial constraints they face, including high costs of finance and inflexible repayment schedules[14]. In particular, it can help reduce the cost of finance through (a) Grant Matching[15]: Grant matching involves providing grants that can be used to leverage additional funding from private investors or financial institutions, thus making it easier for women entrepreneurs to access loans at lower interest rates, as lenders perceive reduced risk when there is a grant backing part of the investment; (b) Guarantees: By offering guarantees to WVSE and WSMEs usually by development organizations, assure lenders that they will cover a portion of the loan in case of default. This reduces the perceived risk for lenders, enabling them to offer loans with more favorable terms—such as lower interest rates or longer repayment periods—thereby lowering the overall cost of finance for women entrepreneurs; and (c) promoting flexible repayment schedules tailored to seasonal cash flows (Tailored financial products, performance-based financing, and providing capacity building support can create an enabling environment that supports sustainable growth and empowers WVSE and WSMEs, especially those in agribusiness.

Addressing Incompatible Aspects of Loan and Cash Flow for Women VSEs and SMEs: Agribusinesses and Women VSE and Women SMEs in the rice postproduction value chain segment often experience significant fluctuations in income due to seasonal cycles. In particular, cash flow is usually high during harvest periods but low or nonexistent during off-seasons, especially in cases of limited funds to purchase paddy and store for off season. In such instances, loan repayment schedules, which are usually fixed monthly payments, do not align with these seasonal cash flows, leading to financial strain. The lack of the necessary financial literacy of WVSE and WSMEs to effectively manage cash flow and understand loan agreements, often lead to poor planning regarding loan repayments, resulting in misalignment between expected cash inflows from business activities and actual repayment schedules. In addition, socio-cultural constrain effective financial planning and management by WVSE and WSMEs, thus further complicating the alignment of cash flows with loan repayments.

These incompatible aspects of loan and cash flow for WVSE and WSMEs can be addressed through targeted

12 https://www.ifc.org/content/dam/ifc/doc/mgrt/202011-new-ifc-discussion-paper.pdf
13 https://mptf.undp.org/sites/default/files/documents/35000/14.08.20_blended_finance_eng.pdf
14 https://chemonics.com/wp-content/uploads/2021/07/Blended-Finance-for-Agricultural-Finance_Final.pdf
15 https://documents1.worldbank.org/curated/en/908381590644619987/pdf/How-Can-Matching-Grants-in-Agriculture-Facilitate-Access-to-Finance-Lessons-Learned-from-World-Bank-Group-s-Experience.pdf

interventions (Amoako-Adu and Eshun, 2018). Targeted interventions can create a more conducive environment for women entrepreneurs in agribusinesses to enhance their ability to manage both loans and cash flows effectively. These could include (a) developing and offering flexible loan products that align with their needs and circumstances; (b) facilitating/offering financial literacy trainings; (c) offering innovative collateral solutions that do not solely rely on traditional assets like land or property. Instead, financial institutions could explore the feasibility of using future earnings from business or inventory as collateral or offering group lending; (d) strengthening advocacy for WVSMEs and WSMEs inclusive financing policies that encourage banks and microfinance institutions to create products tailored specifically for women-led businesses in agriculture; and (e) collaboration between financial institutions and development organizations can help design programs that address the specific needs of women-led VSEs and SMEs in agribusinesses. Such partnerships could focus on providing technical assistance alongside financial support.

Address Misalignment of Repayment Plans and the Needs of WVSE and WSMEs: Loan repayment schedules and cashflow patterns misalign for women very small and small and medium Enterprises in general and in agribusinesses in particular. This is due to the seasonal nature of agribusinesses. WVSE and WSMEs in the rice postharvest segment depend on good harvest for parboiling, milling and sales, thus the inherent seasonal nature of farming cycles that are influenced by climatic condition affects incomes and cashflows, thus loan products that does not take these peculiarities, tend to be misaligned with repayment ability of WVSE and WSMEs. Financial institutions and policy makers should develop appropriate and targeted financial products that take into integrate the specific needs of WVSE and WSMEs in agribusinesses. The products could provide flexible repayment schedules consistent and aligned with agricultural cycles, enhancing access to credit through innovative collateral solutions, improving financial literacy programs targeted at women entrepreneurs, and fostering supportive networks that empower women in business.



4.1.2 Build the Capacity of Financial Institutions to Support WVSEs and WSMEs

Although commercial banks and other formal financial institutions are providing loans to SMEs in other sectors, they consider lending to entrepreneurs in agricultural value chains, especially WVSEs and WSMEs, risky, despite the growing importance of this sector in the economies of West African countries. The perception of WVSEs and WSMEs as risky is due to their informality and, sometimes, their location (rural areas). It is clear from the country reports that the majority of financial institutions have limited knowledge and experience in lending to agriculture enterprises and agribusinesses. In fact, most do not have dedicated desks or departments focusing on agriculture enterprise and agribusiness lending, despite the importance of agriculture in the economies of these countries.

Financial institutions can be capacitated or encouraged to develop specific financial products that meet the unique needs of WVSE and WSMEs and the environment in which they operate. In this regard, financial institutions can be supported in as follows[16]:

1. Better understanding of the needs of WVSE and WSMEs in rice postproduction value chain[17]. This can be done through a needs assessment (the needs, challenges and opportunities) and using the results

16 https://www.snrd-africa.net/wp-content/uploads/2024/02/Accesst2Finance_Final.pdf
17 https://ioe.ifad.org/documents/38714182/41183701/EvSynthesis_Overview_for_web2.pdf/5a779b97-10ab-b5cf-460b-43494d994d23

from the study to guide the development of relevant agricultural finance products. Needs assessment can be conducted through surveys, focus groups interviews, and key informants' interviews. The needs assessment results can be used by financial institutions to identify the needs and challenges of WVSEs and WSMEs.

2. Identification of product development strategies: Using the results from the needs assessment and needs are better understood, financial institutions can adopt several strategies to develop relevant agricultural finance products. Possible relevant products can include (a) Microfinance products (usually smaller loan amounts with flexible repayment terms that align with agricultural cycles) specifically designed for women in postproduction rice value chain segment; (b) Group Lending Models (lending to groups 10-15 per group), encourages peer accountability and collective responsibility, thus reduces risk for lenders while fostering group support among women entrepreneurs; (c) Business Insurance Products (products that protect against machine failure, protection against breakdowns for mechanical equipment like rice mills and equipment) can provide women farmers with a safety net, making them more willing to take on loans, protect against breakdowns for mechanical equipment like rice mills and equipment; (c) Savings Scheme (product that encourage savings) help build capital over time, which can then be leveraged for larger investments or as collateral for loans; and (d) Digital Financial Services[18] (offering digital banking solutions) to expand and enhance access for women who may not have easy access to physical bank branches, particularly in rural areas, where mobile money platforms have proven to be effective for small businesses.

3. Training and capacity building of financial institutions could be provided as technical assistance, with financial support from development partners. Capacity strengthening should target senior management as well as loan officers of financial institutions. The capacity strengthening effort at the senior-management level should improve knowledge on the business opportunities and needs of WVSEs and WSMEs in agriculture enterprises and agribusinesses. This should be to help the senior management develop special loan products for this category of entrepreneurs. At the level of loan officers, they should be capacitated to evaluate the needs of WVSEs and WSMEs for specific agricultural value chains.

4.1.3 Improve Women's Access to Property and Collaterals

IFC estimates that there is a USD 42 billion financing gap for African women across business value chains, including USD 15.6 billion in agriculture alone, largely due to finance, capacity, and the business environment.
As WVSEs and WSMEs have limited or no access to collaterals, thus inhibiting access to bank loans that require them, legal means of improving women's ownership of such productive resources should be legislated. The lack of collateral is a major hindrance for WVSEs and WSMEs to access finance from formal sources. Therefore, countries should develop and enforce legal and regulatory frameworks that support women's legal access to and control of property. As access to and ownership of collateral is a prerequisite for bank loans, addressing this gap is a major step towards formalizing and growing WVSEs and WSMEs. Where such legal instruments and regulatory frameworks exist, effective and proactive information dissemination systems should be put in place to effectively inform women entrepreneurs.

4.2 Addressing Non-Financial Constraints

4.2.1 Develop Country-specific Diagnostics for WVSEs and WSME

Although there are programs supporting women entrepreneurs in general in West Africa, there is no dedicated value-chain financing that targets women alone. In addition, the context of WVSEs and WSMEs differs by country and value chain targeted. In this respect, countries should develop strategies for enhancing WVSE and WSME access to finance. This should be based on a detailed diagnostic of the gaps in WVSE and WSME access to finance and for targeted value chains, as well as the overall business environment in the country. The diagnostics should review existing laws and regulations relating to financing SMEs, the quality of the financial infrastructure, and their responsiveness to the needs of women entrepreneurs. Using the results from the

18 https://documents1.worldbank.org/curated/en/222361617683660582/pdf/Women-in-Agriculture-Using-Digital-Financial-Services-Lessons-Learned-from-Technical-Assistance-Support-to-DigiFarm-Fenix-and-myAgro.pdf



diagnostics, support and incentives should be provided to encourage WVSEs and WSMEs to formalize. Such support could include simplifying the procedures and reducing the time and cost needed to register, reducing taxes, and simplifying tax laws and payment procedures.

4.2.2 Support Women's Access to Motorized Machines and Equipment

Improving access to finance for WVSEs and WSMEs will not only reduce drudgery on women, which has health implications, but will also reduce postharvest losses, and improve rice quality and marketability, thus improving incomes. This includes lending to WVSEs and WSMEs to acquire motorized rice reapers, a motorized machine which cuts rice straws at harvest for threshing; acquire GEM parboiling technology, non-motorized equipment which enhances grain quality and improves energy efficiency; and loans to acquire improved rice milling machines and equipment.

4.2.3 Capacity Building for Financial Institutions in Islamic Banking Services

The Islamic banking system differs from its conventional counterpart. While the conventional system is debt-based, Islamic banking is asset-based and emphasizes real transactions over debt. According to Siddiqi (2007), 'mere debt creation does not increase the net wealth of society, as every addition to social wealth through it is canceled by deduction of a similar amount of wealth owed'. Profit sharing in the Islamic financial system addresses the Islamic ideals of social justice because both the entrepreneur and the lender bear the risk of the investment (Dhumale and Sapcanin, 1999; Siddiqi, 1983).

WVSEs and WSMEs to acquire motorized rice reapers, a motorized machine which cuts rice straws at harvest for threshing; acquire GEM parboiling technology, non-motorized equipment which enhances grain quality and improves energy efficiency; and loans to acquire improved rice milling machines and equipment.

With respect to Islamic banking services, the study on rice postproduction value chain conducted in Guinea, Niger, and Senegal noted that some banks are offering Islamic banking services in those countries.

Capacity building for financial institutions in Islamic banking services encompasses a comprehensive strategy that integrates education on Shari'a compliance, technical skill enhancement, regulatory knowledge, community engagement, continuous professional development, and robust monitoring mechanisms (Durojaye, 2023). This holistic approach ensures that institutions are well-equipped to meet the demands of modern finance while adhering strictly to Islamic principles.

Annex 1: Largest Obstacles Faced by Enterprises in West Africa

Biggest Obstacle to Enterprises

Countries and Data Source	Guinea 2016	Mali 2016	Senegal 2014	S/Leone 2017
Access to finance	6.5	20	39	39.7
Access to land	0	1	4	12.4
Business licenses and permits	0	0.2		0
Corruption	5.7	15.6	4	9
Courts	0.3	1.1	n.a	0
Crime, theft and disorder	9.7	1.2	2	4
Customs and trade regulations	10	3.1	5	2.5
Electricity	4.4	7.5	9	11.8
Inadequately educated workforce	1.2	0.7	1	0
Labor regulations	0.3	3.5	n.a	1.2
Political instability	38.5	23	n.a	2.1
Practices of the informal sector	3	16.3	25	3.2
Tax administration	3.4	1.9	5	2.1
Tax rates	9.7	0.6	4	9.3
Transportation	9.7	4.2	n.a	1.6

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